A fixed finite C1 testing family and an evaluated rational local pencil

Constructive extraction from the complete Q222 source identities

23 October 2026

Abstract

The twelve-root C1 identity admits a source-independent finite testing family on an explicit source box, with no upper bound on $H = h/\sigma$. A fixed matrix reserve and two algebraic gate variables replace the point-dependent TF1 radicals. A complementary six-root identity treats $\lambda \le 0$. Keeping both independent state normals and the complete retained squares gives an evaluated rational source-cone pencil of total order 669, with 120 blocks, maximum block order 12, and 2694 auxiliaries. Both directions of its representation of the specified *local certificate cone* are proved. The construction does not give an exact lift of the full eight-coordinate $\mathcal{Q}_{222}$: global source and independent testing coverage, including the other C1 and C2 boundaries, remains to be extracted.

1 The result and its exact scope

Write $\mathbf{c} = (d, m, C, A, D, \alpha, \beta)$ for the seven canonical coefficients of the reciprocal parent $F_{\mathbf{c}}$ in Section 2. Let $\mathcal{K}_{\mathrm{POS}}$ and $\mathcal{K}_{\mathrm{NEG}}$ be the two fixed Gram cones constructed below on the state charts $W \ge 0, Z \ge 0$ and $W \ge 0, Z \le 0$, respectively. The names refer to state signs, not to the sign of λ. Let $\mathcal{K}_{\mathrm{SIX}}$ be the fixed six-root certificate cone of Section 7. Define

$$\mathcal{K}_{669} = (\mathcal{K}_{\mathrm{POS}} \cap \mathcal{K}_{\mathrm{NEG}}) + \mathcal{K}_{\mathrm{SIX}}.$$

These cones are defined by identities in the full polynomial $F_{\mathbf{c}}$ and unrestricted positive semidefinite Gram blocks on the printed testing vectors. They do not impose the nonlinear stationary source formulas on their visible coordinates.

Theorem 1 (Explicit local fixed-family admission)**.** *Use the native rational stationary source (P1), with $d = 1$, and impose*

$$0 < x, \sigma \le 2^{-30}, \qquad 0 < z \le 1, \qquad 0 < b, h < 1, \quad r = bz < 1, \quad E, M, J > 0, \quad D, \beta \ge 0. \tag{D}$$

Then $\mathbf{c} \in \mathcal{K}_{669}$. There is no restriction on the sign of λ and no upper bound on $H = h/\sigma$. Every $\mathbf{c} \in \mathcal{K}_{669}$ satisfies

$$F_{\mathbf{c}}(u, w; W, X, Y, Z) \ge 0 \quad \text{for } |u| \le 2^{-24},\ |w| \le 2^{-12}, \quad (W, X, Y, Z) \in \mathbb{R}^4. \tag{T}$$

The statement extends to complex states by real and imaginary splitting. All testing axes and zero state coordinates are literal specializations. Both permitted source faces $D = 0$ and $\beta = 0$ are retained. Positive multiples of an admitted source are admitted as well.

Theorem 2 (Evaluated local cone representation)**.** *The supplied rational matrix file defines a homogeneous affine pencil $L_{669}(\mathbf{c}, y)$ such that, for every real $\mathbf{c}$,*

$$\boxed{\mathbf{c} \in \mathcal{K}_{669} \iff \exists\, y \in \mathbb{R}^{2694} :\ L_{669}(\mathbf{c}, y) \succeq 0.}$$

Its total order is 669, its 120 diagonal blocks have order at most 12, and its upper triangle has 7716 nonzero rational coefficient records. There are no unevaluated matrix entries.

Fixed cone	Blocks	Total order	Largest block	Auxiliaries
$\mathcal{K}_{\mathrm{POS}}$	52	268	12	1087
$\mathcal{K}_{\mathrm{NEG}}$	49	240	11	934
$\mathcal{K}_{\mathrm{POS}} \cap \mathcal{K}_{\mathrm{NEG}}$	101	508	12	2021
$\mathcal{K}_{\mathrm{SIX}}$	19	161	10	666
$\mathcal{K}_{669}$	120	669	12	2694

The matrix file is `C1_UNIFORM_LOCAL_ORDER669.coo.json`. The last line includes seven source-splitting variables. Theorems 1–2 concern the independent testing box (T). Feasibility of this pencil does not prove validity on the other quantum testing regions. In particular this pencil cannot yet be polarized as an exact full-$\mathcal{Q}_{222}$ lift.

Proof route. Section 2 states the complete source and parent identities. Section 3 extracts the twelve-root Gram and every retained square. Section 4 supplies numerical source bounds. Section 5 replaces TF1 by a fixed algebraic family. Section 7 handles the complementary circuit sign. Section 8 proves both local cone inclusions and specifies the evaluated files. The final section isolates the global obligation that remains.

2 Source formulas and the complete parent

The independent source parameters are x, σ, b, z, h. Source parameters and testing coordinates are independent throughout. Set

$$r = bz, \quad Q_b = 1-b, \quad \theta = 1-h, \quad H = h/\sigma, \quad e = xH, \quad E = 1-e, \quad K = 1+x,$$
$$R_s = 1+xr, \qquad M = R_s - xz\theta, \qquad J = 1+xh-\sigma(1-r),$$
$$P = \sigma + K + \sigma z(1+bx+2xh) + \sigma b x^2 h z^2 + x^2 h\theta z.$$

The native domain has $x, \sigma, z > 0$, $0 < b, r, h < 1$, $E, M, J > 0$, and $D, \beta \geq 0$. Put

$$p = \frac{z(1+xh)E + (1-r)(1+xhz)}{2P},$$
$$l = \frac{(1+\sigma r)\{eR_s(1+\sigma z) + ME\}}{2P}, \qquad \kappa = \frac{K(1+\sigma r)(1+\sigma z)}{2MP}.$$

The canonical source coefficients, with $d = 1$, are

$$m = \kappa x^2 z H, \qquad C = \frac{Kz(1+\sigma r)E}{2(1-r)P}, \qquad A = p/r, \qquad \alpha = l/(zQ_b),$$
$$D = \frac{K(1+\sigma z)(1+\sigma r) - JP}{2JP}, \qquad \beta = \frac{K(1+\sigma r)(1+xhz) - \theta P}{2\theta P}. \tag{P1}$$

In particular $a = Az = p/b$, $\ell = \alpha z = l/Q_b$, $k = \sqrt{x} > 0$, and $q = \sqrt{z} > 0$. The symbols A and α here are source coefficients; the capital testing gates A_t, B_t used in some displays below are unrelated.

Two exact source identities will provide the bounds. Define

$$H_0 = 1 + zQ_b, \qquad B_s = P + xz(J - \sigma),$$
$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + Ke^2\}.$$

Direct multiplication of (P1) gives

$$2\theta P\beta = hB_s - \sigma H_0, \qquad 2P(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{BE}$$

For an arbitrary canonical vector $\mathbf{c}$ define the full four-state parent

$$\begin{aligned} F_{\mathbf{c}} = {} & 4m(W - wX + uY - uwZ)^2 + 4C(1+w^2)(uW - Y)^2 \\ & + 4A(1+u^2)(1+w^2)Y^2 + (4D+2d)(uwW + uX - wY - Z)^2 \\ & + (4\alpha + 2d)(1+u^2)(Y - wZ)^2 + (4\beta + 2d)(1+w^2)(X + uZ)^2 \\ & - 2d(1+u^2)(1+w^2)Z^2. \end{aligned} \tag{P4}$$

Thus $F_{\mathbf{c}}$ is linear in $\mathbf{c}$. Write $F_w = F_{\mathbf{c}}$ for the source (P1), with $d = 1$.

Here are every source coefficient and every auxiliary form needed in the construction:

$$g = h - Q_b, \quad \nu = (2D - 2\beta\theta + h)/\sigma,$$
$$\lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z, \qquad \lambda_0 = \lambda - 2H, \qquad K_A = 4\kappa H - \lambda,$$
$$B_x = 8\kappa e, \qquad A_y = 8(p-l) + 4z\{(1+2m)g + 2Dh + 2p\sigma\},$$
$$R_x = -4\{2Db + (1+2m)g - 2\beta\theta\}, \qquad R_0 = -4\{2(C+D)(r-1) + 2p + r - 1\}, \tag{SP3}$$
$$\chi = 8(pb + lQ_b) - 4h + 4eh(2D+1) - 8Dh$$
$$+4z\{4Dbh + 4pb\sigma + (1+2m)g^2 + 2\beta\theta^2\},$$
$$k_1 = 2(1+2m)g^2 + 8Dhb + 8pb\sigma + 4\beta\theta^2.$$

Set $t = u^2$, $\xi_w = w^2$ and

$$N_x = X + uhZ, \quad N_y = (Y - bwZ)/q, \quad \Gamma = \sigma Z - uwW, \quad R = wZ/q - uqW,$$
$$A_\theta = q(t-x)W, \quad P_\beta = uZ - (x/\sigma)wW, \quad X_e = X + ewW, \quad V = uw(t-x)W,$$
$$J_C = N_y + bR, \qquad J_\ell = N_y - Q_b R, \qquad R_\gamma = R_0 - 8pt.$$

Define

$$\begin{aligned}
E_{yy} &= 2z\{(1+2m)t + (1+2D)\xi_w\},\\
E_{RR} &= z\{8bD + 8bp\sigma + 4b\sigma(2C+2D+1)(r-1) + k_1 t + 2b^2(2D+1)\xi_w\},\\
E_{yR} &= 4z\{t[2Dh + 2p\sigma + (1+2m)g] + b(2D+1)\xi_w\\
&\qquad + \sigma(2C+1)(r-1) + 2D[1+\sigma(r-1)] + 2p\sigma\},\\
\mathcal{E}_+ &= 4\beta(1+w^2)N_x^2 + 2(2D+1)(uX_e)^2 + 4bpzt\Gamma^2\\
&\quad + 4a[(1+t)(1+w^2) - 1]J_C^2 + 4\ell t J_\ell^2 + 4Czw^2 J_C^2,\\
\mathscr{S} &= \lambda\sigma P_\beta^2 + (\lambda x/\sigma)(t-x)(w^2 - \sigma z)W^2.
\end{aligned}$$

The complete polynomial identity is

$$\begin{aligned}
F_w = \mathcal{E}_+ &+ [2(1+w^2) + 4mw^2]N_x^2 + [4a + 2z(2C+1)]J_C^2 + 4\ell J_\ell^2\\
&+ E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_y R + (4D/\sigma^2)\Gamma^2\\
&- 4\nu N_x(u\Gamma) + R_\gamma(N_y + bR)(qw\Gamma)\\
&- 4uqw(2m + 2D + 1)N_x N_y + uqwR_x N_x R\\
&+ qwB_x N_x A_\theta + uA_y N_y A_\theta + K_A A_\theta^2 + \chi u A_\theta R + \mathscr{S}.
\end{aligned} \tag{SP6}$$

To verify (SP6), substitute the printed linear forms and collect the ten quadratic state monomials, then the powers of u, w. Substitute (P1) and clear the positive source denominators; each coefficient is zero. This is an algebraic identity, with both distinct Γ couplings displayed explicitly. The accompanying native-parent verifier performs this generic rational comparison, not numerical evaluation or a determinant comparison.

The two transformations

$$(u, Y, Z) \mapsto (-u, -Y, -Z), \qquad (w, X, Z) \mapsto (-w, -X, -Z) \tag{SG}$$

preserve (P4) for every $\mathbf{c}$. It therefore suffices to construct the certificate at $u, w \geq 0$. An overall state sign permits $W \geq 0$. Both signs of Z will be included.

3 The complete twelve-root extraction

This section gives the complete polynomial identity behind the finite testing family. All lower-case source parameters in this section are in the coefficient field. The variables u, w, W, X, Y, Z are independent testing variables. Put $q = \sqrt{z}$, $x = k^2$, $Q_b = 1 - b$, and use the native stationary C1 source of (P1). In particular

$$a = p/b, \quad \ell = l/Q_b, \quad e = xh/\sigma, \quad \rho = \lambda x/\sigma, \quad L_0 = \lambda\sigma, \quad q_0 = \lambda k, \quad d_G = 4D/\sigma^2, \quad S = d_G + \rho.$$

Here $\lambda > 0$. Set $a_0 = k/\sigma$, $c = \max(0, 1 - K_A)$ and $K_+ = K_A + c \geq 1$. No inverse of D, β, or c is used.

The polynomial is the complete parent (P4), with all four state coordinates independent. Its full source definitions are (P1) and (SP3).

3.1 The twelve principal roots

Define

$$N_x = X + uhZ, \qquad N_y = (Y - bwZ)/q, \qquad R = wZ/q - uqW,$$
$$\Gamma = \sigma Z - uwW, \qquad G_* = \sqrt{d_G}\,\Gamma,$$
$$A_\theta = q(u^2 - k^2)W, \qquad T_\theta = kq(u-k)W,$$
$$U_* = \sqrt{q_0}\,q\sqrt{u}(u-k)W, \qquad K_* = \sqrt{q_0 k}\,q(u-k)W.$$

For $u, w, W, Z \geq 0$ put

$$B_1 = \sqrt{L_0}\, uZ, \quad B_2 = \sqrt{\rho}\, uwW, \quad B_3 = \sqrt{\lambda}\, k^2 qW, \quad B_4 = \sqrt{\lambda}\, kwZ/q,$$
$$G = \sqrt{\lambda}\, k\sqrt{uwWZ}, \qquad d_i = B_i - G.$$

Thus

$$\psi = (N_x, N_y, R, G_*, A_\theta, T_\theta, d_1, d_2, d_3, d_4, U_*, K_*)^{\mathsf{T}} = A_{\mathrm{src}}\phi,$$

where A_{src} has only source entries and

$$\phi = (X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^{3/2}W, u^{1/2}W, \sqrt{uwWZ})^{\mathsf{T}}.$$

This proves the claimed separation-rank bound twelve. In particular, the fourth-root circuit does not add an independent principal testing function: $B_1B_2B_3B_4 = \lambda^2x^2u^2w^2W^2Z^2$ gives the printed formula for G exactly.

If $W \geq 0$ and $Z \leq 0$, retain the same signed B_i and put $d_i = B_i$. The twelfth column of A_{src} is then zero. The full matrix is printed in `C1_ROOT_MATRIX.json` for both signs.

3.2 Both state-normal conductors and the exact Gram

Write $J_C = N_y + bR$ and $J_\ell = N_y - Q_bR$. The two linear forms

$$\mathfrak{U} = \frac{\sqrt{d_G}uG_* + \sqrt{\rho}\{k(d_1 - d_2) - (u-k)(d_2 - d_3)\} - \rho\sqrt{\sigma}T_\theta}{S},$$
$$\mathfrak{V}_0 = \frac{R_0\sqrt{d_G}qwG_* + R_0\rho\sigma q^2R - R_0\sqrt{\rho}(qw + q^2\sqrt{\sigma})(d_2 - d_3)}{S}$$
$$+ \frac{R_0q_0q^2 + (R_0\sqrt{\lambda})\sqrt{\rho}qw}{S}T_\theta, \qquad \mathfrak{V} = \mathfrak{V}_0 - 8pqwu\mathfrak{U}$$

satisfy, exactly,

$$\mathfrak{U} = u\Gamma, \qquad \mathfrak{V} = (R_0 - 8pu^2)qw\Gamma.$$

Both complete normal terms are therefore retained. These formulas are also literal when $D = 0$.

Let

$$E^0_{RR} = z\{8bD + 8bp\sigma + 4b\sigma(2C + 2D + 1)(r - 1)\},$$

and let $E_{yR}, \nu, R_0, R_x, B_x, A_y, \chi$ have the explicit source definitions in (SP3). Put

$$\varepsilon_C = 2/\lambda + u + k, \qquad \varepsilon_G = 2a_0u + (u + k)/L_0.$$

The exact principal Gram is the following quadratic form in the twelve formal coordinates ψ:

$$\psi^{\mathsf{T}}\mathcal{Q}\psi = 2N_x^2 + J_C^2 + J_\ell^2 + G_*^2 + \frac{K_+}{2}(A_\theta^2 + T_\theta^2) + \frac{1}{2}\sum_{i=1}^{4} d_i^2 + U_*^2 + \frac{1}{2}K_*^2$$
$$+ \left(E^0_{RR} - \frac{\lambda x}{2} - \frac{cd_Gu^2}{S}\right)R^2 + E_{yR}N_yR - 4\nu N_x\mathfrak{U} + J_C\mathfrak{V}$$
$$- 4uqw(2m + 2D + 1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta + \chi uA_\theta R$$
$$+ \sqrt{q_0u}\,U_*R - \frac{c}{S}(\sqrt{a_0u}\,U_* + \sqrt{a_0k}\,K_*)^2 - \frac{c\varepsilon_G}{S}G_*^2 - \frac{cd_G\varepsilon_C}{S}\sum_{i=1}^{4} d_i^2.$$

In $s = \sqrt{u}, w$, this matrix has just the testing monomials

$$1, \quad w, \quad w^2, \quad s, \quad s^2, \quad s^2w, \quad s^4, \quad s^4w.$$

The eight evaluated symmetric coefficient matrices, comprising 104 stored upper-triangle entries, are in `C1_PRINCIPAL_GRAM.json`.

3.3 The nineteen elementary retained roots

Define

$$X_e = X + ewW, \qquad F_A = u(Z - a_0 wW),$$
$$k_1 = 2(1+2m)(h-Q_b)^2 + 8Dhb + 8pb\sigma + 4\beta(1-h)^2.$$

The following nineteen roots are retained without any mixed square root. Every displayed radical is purely source-side.

$$2\sqrt{\beta}N_x, \quad 2\sqrt{\beta}wN_x, \quad \sqrt{2(2D+1)}uX_e, \quad 2\sqrt{bp}\,qu\Gamma,$$
$$2\sqrt{a}\,uJ_C, \quad 2\sqrt{a}\,wJ_C, \quad 2\sqrt{a}\,uwJ_C, \quad 2\sqrt{\ell}\,uJ_\ell, \quad 2\sqrt{C}\,qwJ_C,$$
$$\sqrt{2+4m}\,wN_x, \quad \sqrt{4a+2z(2C+1)-1}\,J_C, \quad \sqrt{4\ell-1}\,J_\ell,$$
$$\sqrt{2(1+2m)}\,quN_y, \quad \sqrt{2(1+2D)}\,qwN_y, \quad \sqrt{k_1}\,quR, \quad \sqrt{2(2D+1)}\,qbwR,$$
$$\sqrt{L_0/2}\,F_A, \quad \sqrt{K_+/2}\,qu(u-k)W, \quad \sqrt{K_+k}\,q\sqrt{u}(u-k)W.$$

The final pair represents $K_+(A_\theta^2 - T_\theta^2)/2$. The native source bounds give $4a > 1$ and $4\ell > 1$, so their radicands are nonnegative. Each root is a source-linear combination of at most three fixed testing monomials.

3.4 Eight angular roots, and the negative-state correction

Let

$$\mathscr{C} = \sum_{i=1}^{4} B_i^2 - 4\lambda xuwWZ, \qquad \Delta_A = \varepsilon_C \mathscr{C} + u^2R^2 + \varepsilon_G\Gamma^2 - A_\theta^2.$$

For $W, Z \geq 0$ put

$$L = \sqrt{q_0\sigma}\,F_A + \frac{k}{\sqrt{q_0\sigma}}\Gamma, \quad K_Z = k^2qW - kwZ/q, \quad H_Z = \sqrt{wWZ}.$$

The elementary angular identity used here is, for $U, V, Z_* \geq 0$ and $a_* > 0$,

$$\Psi = 1 + U^2V^2 + U^2Z_*^2 + V^2Z_*^2 - 4UVZ_*, \quad T_* = U^2 - 1, \quad Q_* = VZ_* - U,$$
$$R_* = Z_* - UV, \qquad G_* = U(V - Z_*),$$

$$\begin{aligned}
&[2 + a_*(U+1)]\Psi + U^2Q_*^2 + [2U + (U+1)/a_*]R_*^2 - T_*^2 \\
&= (T_* + UQ_*)^2 + 2Q_*^2 + (U+1)(\sqrt{a_*}G_* - R_*/\sqrt{a_*})^2 \\
&\quad + a_*(U+1)(1 - VZ_*)^2 + 2a_*(U+1)VZ_*(U-1)^2.
\end{aligned} \tag{US}$$

Multiplication verifies it. For $W \neq 0$, substitute

$$U = u/k, \quad V = w/(\sqrt{\sigma}q), \quad Z_* = \sqrt{\sigma}Z/(kqW), \quad a_* = q_0$$

and multiply by $(k^2qW)^2$. The resulting formulas below no longer divide by W or Z, so their polynomial identity includes those zero coordinates.

Then Δ_A is the sum of the squares of exactly these eight roots:

$$A_\theta + uR, \qquad \sqrt{2}kR, \qquad L, \qquad \sqrt{u/k}\,L,$$
$$\sqrt{q_0}K_Z, \qquad \sqrt{q_0u/k}K_Z, \qquad \sqrt{2q_0k}(u-k)H_Z, \qquad \sqrt{2q_0u}(u-k)H_Z.$$

This follows by multiplying out (US); the displayed formulas no longer divide by a state coordinate. Multiply these roots by the pure-source factor $\sqrt{cd_G/S}$.

If $W \geq 0, Z \leq 0$, first use the eight formulas with $|Z|$ in place of Z, and append the three roots

$$2k\sqrt{4 + 2q_0 + q_0^{-1}}\sqrt{u}H_{|Z|}, \quad 2\sqrt{k}\sqrt{2q_0 + 2 + q_0^{-1}}uH_{|Z|}, \quad 2u\sqrt{u}H_{|Z|}.$$

Their squared sum is exactly the increase in Δ_A when the positive state coordinate is replaced by its negative. This proves the eleven-root formula on the negative-state chart.

3.5 The three circuit roots and their fixed testing span

On the positive-state chart set

$$a_t = u^{1/4}, \quad b_t = w^{1/4}, \quad \omega = W^{1/4}, \quad \zeta = Z^{1/4}, \quad \gamma = \sqrt{\sqrt{\lambda}\,k}.$$

The six testing monomials are

$$f_1 = a_t^3 b_t \omega \zeta^3, \quad f_2 = a_t^3 b_t^3 \omega^3 \zeta, \quad f_3 = a_t b_t \omega^3 \zeta, \quad f_4 = a_t b_t^3 \omega \zeta^3,$$
$$f_5 = a_t^3 b_t^2 \omega^2 \zeta^2, \qquad f_6 = a_t b_t^2 \omega^2 \zeta^2.$$

The retained $\mathscr{C}_+ = \mathscr{C} - \sum d_i^2$ is the sum of the squares of

$$\sqrt{2}\gamma(L_0^{1/4} f_1 - \rho^{1/4} f_2),$$
$$\sqrt{2}\gamma\{(\sqrt{\lambda}k^2 q)^{1/2} f_3 - (\sqrt{\lambda}k/q)^{1/2} f_4\},$$
$$2\gamma\{(\lambda k)^{1/4} f_5 - (\lambda k^3)^{1/4} f_6\}.$$

On the negative-state chart there is instead the single root $2\sqrt{\lambda}k\sqrt{uwW|Z|}$.

The common scalar coefficient retained in the parent is

$$q_C = \frac{1}{2} - \frac{c d_G}{S}(2/\lambda + k + u).$$

On $0 \le u \le \delta^2$, with $u = \delta^2 A_t^8$ in the scaled testing parametrization, write $A_C = c d_G \delta^2 / S$ and $q_C^0 = \frac{1}{2} - c d_G (2/\lambda + k)/S$. Exactly

$$q_C = (q_C^0 - A_C) + A_C(1 - A_t^8), \qquad 1 - A_t^8 = \xi^2(1 + A_t^4), \quad \xi^2 = 1 - A_t^4.$$

Thus the three fixed multipliers $1, \xi, \xi A_t^2$ suffice for each circuit root. There is no point-dependent choice of a radical or a bounding constant.

3.6 Exact assembly and boundary specialization

The principal quadratic form, the nineteen elementary squares, the weighted angular squares, and $q_C \mathscr{C}_+$ sum to F_w. This identity was independently verified first against the complete SP6 expression and then against the original P4 parent after the generic rational P1 substitution. In the latter check all seventeen nonzero pre-substitution coefficients in the independent polynomial variables W, X, Y, Z, u, w vanish identically.

All final testing functions are polynomial in quarter-angle and quarter-state coordinates. Consequently $u = 0, w = 0, W = 0, Z = 0$ are literal specializations, with no limiting SOS argument. The signs of the quarter-state parameters do not matter: each circuit root has one fixed parity in each parameter, so its square is invariant. The testing sign gauges $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and $(w, X, Z) \mapsto (-w, -X, -Z)$ restore the other angle signs. An overall state sign makes $W \ge 0$; the two state charts then cover all real states. Real and imaginary parts restore complex states.

The complete retained coefficient matrices are in `C1_RETAINED_ROOTS.json`: the positive-state chart has 33 distinct retained testing monomials and 30 roots before the circuit coefficient is split; the negative-state chart has 27 monomials and 31 roots. Every retained root has separation rank at most three. These are local certificates for the printed source and testing domains; none of these identities asserts that this one chart is an exhaustive global Q222 testing family.

4 A fully evaluated source box, with arbitrary H

The pointwise estimates in the uniform C1 proof can be replaced by numerical inequalities. Set

$$\epsilon = 2^{-30}, \qquad \delta = 2^{-12}.$$

Use the native source formulas (P1), the definitions (SP3), and impose

$$0 < x, \sigma \le \epsilon, \quad 0 < z \le 1, \quad 0 < b, h < 1, \quad r = bz < 1, \quad E, M, J > 0, \quad D, \beta \ge 0. \tag{1}$$

There is no lower bound on $z, b, 1-b, h, 1-h$, and no upper bound on $H = h/\sigma$. The literal faces $D = 0$ and $\beta = 0$ are admitted. The opening variables x, σ, z remain positive in this chart; a literal zero opening is not obtained by cancelling a zero factor.

Here and below every displayed inequality holds in the source real closed field. Put $k = \sqrt{x}$, $q = \sqrt{z}$, and $e = xh/\sigma$. The following estimates will be deliberately conservative.

$$1 \le P \le 1 + 4\epsilon, \quad 1 - \epsilon \le J, M \le 1 + \epsilon, \quad 1 \le B_s \le 1 + 6\epsilon, \quad 1 \le H_0 \le 2, \tag{2}$$

$$0 < e \le 6\epsilon, \quad |\mathcal{H} - H_0| \le 32\epsilon, \quad 0 \le D \le 3\sigma^2, \quad 0 \le \beta(1-h) \le h, \tag{3}$$

$$\tfrac{1}{3} \le p, l, \kappa \le 2, \quad \kappa \le 1, \quad 0 < m \le 6\epsilon^2, \quad |C(r-1)| \le 1, \tag{4}$$

$$|\nu| \le 4, \quad |\lambda_0| \le 5, \quad |K_A - \{2(1-h)H_0 - 2h(1-r)\}| \le 1024\epsilon, \tag{5}$$

$$|R_0| \le 40(x+\sigma), \quad |B_x| \le 48\epsilon, \quad |A_y|, |\chi| \le 64, \quad |R_x| \le 24. \tag{6}$$

Proof of the bounds. For the proof one may first replace 2^{-30} by any positive real $\epsilon \le 2^{-12}$. The formula for P is a sum of nonnegative terms. Its nonconstant part is at most $x + 2\sigma + 3x\sigma + x^2\sigma + x^2 \le 4\epsilon$. The formulas for J, M, B_s give (2) directly; in particular $J - \sigma > 0$.

Initially $0 < e < 1$ follows from $E > 0$. Dropping the negative term in $\mathcal{H}$ gives $\mathcal{H} \le 4$. The source identities

$$2(1-h)P\beta = hB_s - \sigma H_0, \qquad 2P\{JD + x(1-h)\beta\} = \sigma^2\mathcal{H}$$

imply

$$eB_s \le xH_0 + \sigma\mathcal{H} \le 2x + 4\sigma.$$

Thus $e \le 6\epsilon$. Expanding $\mathcal{H} - H_0$ and taking absolute values gives

$$|\mathcal{H} - H_0| \le 5x + 3e + 2e^2 \le 32\epsilon.$$

The second source identity, with both summands nonnegative, gives $D \le 3\sigma^2$. The first gives $\beta(1-h) \le hB_s/(2P) \le h$. The numerator of p is at least $H_0 - e \ge 1 - 6\epsilon$; the numerator of l is at least $M(1-e)$. These inequalities, together with (2), give the lower bounds $p, l \ge 1/3$. The corresponding upper bounds and $1/3 \le \kappa \le 1$ follow by bounding each positive numerator factor. Now $m = \kappa xze$ and the pole-free expression for $C(1-r)$ prove the rest of (4).

Use the exact identities

$$\begin{aligned}\nu &= 2D/\sigma + \{H_0 - ez(J-\sigma)\}/P,\\ \lambda_0 &= 2e(1-h)z(J-\sigma)/P - 2(1-h)H_0/P + 8Dh/\sigma - 4bp\sigma z.\end{aligned}$$

They imply $|\nu| \le 4$, $|\lambda_0| \le 5$, and

$$|\lambda_0 + 2(1-h)H_0| \le 72\epsilon.$$

To bound K_A uniformly in H, write

$$N_\kappa = (1+x)(1+\sigma r)(1+\sigma z) - MP = xF_x + \sigma F_\sigma.$$

Here $F_x = (N_\kappa - N_\kappa|_{x=0})/x$ is a polynomial with sum of absolute coefficients 37 after $r = bz$, and $F_\sigma = r - 1 + \sigma rz$. All its arguments belong to $[0,1]$. The exact pole removal is

$$H(2\kappa - 1) = \frac{eF_x + hF_\sigma}{MP}.$$

Since $|(MP)^{-1} - 1| \le 12\epsilon$ and $(MP)^{-1} \le 2$,

$$|H(2\kappa - 1) + h(1-r)| \le 458\epsilon.$$

Combining this with $K_A = 2H(2\kappa - 1) - \lambda_0$ proves (5), since $2 \cdot 458 + 72 < 1024$.

For R_0, eliminate the apparent C pole explicitly:

$$\begin{aligned}-PR_0/4 = {}& z(1-e)\{1 + xh - (1+x)(1+\sigma r)\}\\ &+ (1-r)\{1 + xhz - P\} - 2DP(1-r).\end{aligned}$$

Taking absolute values gives $|R_0| \le 40(x+\sigma)$. The remaining bounds in (6) follow immediately from the definitions (SP3), $|h + b - 1| \le 1$, and (3)–(4).

The positive-λ branch. Suppose additionally $\lambda > 0$ and put

$$S = d_G + \rho, \quad d_G = 4D/\sigma^2, \quad \rho = \lambda x/\sigma, \quad L_0 = \lambda\sigma, \quad q_0 = \lambda k.$$

The exact identity

$$\begin{aligned} S = 2\mathcal{H}/P + d_G(1 - J + 2hx) + 2eH_0/P \\ - 2e^2 z(J - \sigma)/P - 4bpxz \end{aligned}$$

and the preceding bounds give

$$1 \le S \le 10, \quad 0 \le d_G, \rho \le 10, \quad 0 < L_0 \le 3, \quad 0 < q_0 \le 6, \quad \lambda x \le 10\epsilon. \tag{7}$$

For example, the negative terms in the expression for S have magnitude at most $144\epsilon^2 + 8\epsilon$; its leading term is at least $2(1 - 32\epsilon)/(1 + 4\epsilon)$. This gives $S \ge 1$. The upper bound follows by dropping the two negative terms and using $d_G \le 12$ before improving it to $d_G \le S \le 10$. Finally $q_0^2 = L_0\rho \le 30$.

The remaining potentially large expression is controlled without bounding λ itself:

$$(x + \sigma)^2 H = xe + 2\sigma e + \sigma h \le 2\epsilon, \qquad (x + \sigma)^2 \le 4\epsilon^2.$$

Since $\lambda \le 2H + 5$, it follows that

$$|R_0\sqrt{\lambda}| \le 100\sqrt{\epsilon}. \tag{8}$$

Let $c = \max(0, 1 - K_A)$ and $a_0 = k/\sigma$. The error bound for K_A gives $0 \le c \le 4$. If $c > 0$, then $h > 1/8$: for $h \le 1/8$ the reference value of K_A is at least $3/2$, and its error is at most $1/4$. On this branch $L_0 = 2h + \sigma\lambda_0 \ge h > 1/8$. Moreover

$$a_0^2(hB_s - \sigma H_0) \le \mathcal{H}, \qquad hB_s - \sigma H_0 \ge 1/16,$$

so $a_0 \le 8$. Consequently

$$c/L_0 \le 32, \quad ca_0 \le 32, \quad c/\lambda \le 32\sigma. \tag{9}$$

When $c = 0$, all terms containing these factors are exactly zero; no positive lower bound for L_0 or upper bound for a_0 is needed.

Bounds on every coefficient matrix. Let $s = \sqrt{u}$, and write the complete principal Gram from the source export as

$$Q(s, w) = Q_0 + \sum_{(i,j)\in\mathcal{A}} C_{ij} s^i w^j, \quad \mathcal{A} = \{(0,1), (0,2), (1,0), (2,0), (2,1), (4,0), (4,1)\}.$$

Its source reference B has N_x entry 2, radial block

$$\begin{pmatrix} 2 & 2b - 1 \\ 2b - 1 & b^2 + (1 - b)^2 \end{pmatrix},$$

Gamma entry 1, two angular entries $(K_A + c)/2$, four circuit entries $1/2$, and final entries $1, 1/2$. There are no other entries. The exact identity

$$\begin{aligned} (N_y + bR)^2 + (N_y - (1 - b)R)^2 = \frac{N_y^2 + R^2}{3} + \frac{5}{3}\left(N_y + \frac{3(2b - 1)}{5}R\right)^2 \\ + \frac{1 + 6b(1 - b)}{15}R^2 \end{aligned}$$

proves $B \succeq \frac{1}{3}I_{12}$ without any choice of whitening matrix. Here $K_A + c \ge 1$ by definition.

For a symmetric matrix use the row norm $\|A\|_\infty = \max_i \sum_j |A_{ij}|$. Substitution of (2)–(9) into the printed matrices yields

$$\|Q_0 - B\|_\infty \le 2048\sqrt{\epsilon}, \qquad \|C_{ij}\|_\infty \le 128 \quad ((i,j) \in \mathcal{A}). \tag{10}$$

These are finite coefficient bounds, not bounds obtained after a test has been selected. To make the entrywise audit transparent, the largest possible s^2 diagonal correction is $c(2a_0 + 1/L_0)/S \le 96$; its adjacent normal coupling is at most $2\sqrt{10}|\nu| < 26$, so its row sum is below 128. The three-entry normal rows in the s^2 and $s^4 w$ matrices are at most $6|\nu|\sqrt{10} < 76$ and $12p\sqrt{10} < 76$, respectively. The s matrix is bounded by $\sqrt{6}/2 + 32\sqrt{k}$.

The w matrix is bounded using (8); all its entries are $O(\sqrt{\epsilon})$ with the explicit constants above. The w^2, s^2w, s^4 rows are bounded by $4, 32, 44$, respectively. The constant correction is bounded by grouping the pole-free $C(r-1)$ terms before taking absolute values; its normal rows are at most $144\sqrt{\epsilon}$, its radial rows at most 612ϵ, and its circuit rows at most $72\sqrt{\epsilon} + 1000\epsilon$. All these are below the bound in (10).

Define the evaluated rational number

$$R_* := 128(2\delta + 2\delta^2 + \delta^3 + \delta^4 + \delta^5) = \frac{563087409156097}{9007199254740992} < \frac{1}{12}.$$

Because $2048\sqrt{\epsilon} = 1/16 < 1/12$, the source matrices

$$Q_0 - R_* I_{12}, \qquad \frac{\delta^{i+j}}{2}(128 I_{12} \pm C_{ij}) \quad ((i,j) \in \mathcal{A}) \tag{11}$$

are positive semidefinite. In fact the first is positive definite with lower bound greater than $1/6$. Thus the fixed testing construction applies throughout (1) with $\lambda > 0$, including both weak source-weight faces.

5 A fixed algebraic compiler for the twelve-root C1 Gram

This section removes the adaptive testing radicals from the principal $\lambda > 0$ C1 Gram on one explicit testing box. The source admission condition is finite and algebraic. It is a sufficient condition, and no claim that it covers every stationary source is made.

Let $\delta = 2^{-12}$, $s = \sqrt{u}$, and write the complete principal Gram as

$$Q(c; s, w) = C_{00}(c) + \sum_{(p,q)\in I} C_{pq}(c) s^p w^q, \qquad I = \{(0,1),(0,2),(1,0),(2,0),(2,1),(4,0),(4,1)\}.$$

Here c denotes the source, rather than the angular compensation scalar; the printed source matrices retain both independent state normals. Their order is twelve. Define the source reference matrix $B(c)$ by the diagonal entries

$$(2,\ 2,\ b^2 + (1-b)^2,\ 1,\ (K_A + c_A)/2,\ (K_A + c_A)/2, 1/2, 1/2, 1/2, 1/2, 1, 1/2)$$

and the single off-diagonal entry $B_{23} = B_{32} = 2b - 1$ in one-based indexing, where $c_A = \max(0, 1 - K_A)$. Its radial block has determinant one and trace at most three for $0 \le b \le 1$. Thus

$$B(c) \succeq \tfrac{1}{3} I_{12}.$$

An elementary exact verification of its radial reserve is

$$(N_y + bR)^2 + (N_y - (1-b)R)^2 = \tfrac{1}{3}(N_y^2 + R^2) + \tfrac{5}{3}\left(N_y + \tfrac{3}{5}(2b-1)R\right)^2 + \tfrac{1+6b(1-b)}{15} R^2.$$

No source-dependent real whitening is used.

For a symmetric source matrix M, put

$$n(M) = \max_i \sum_j |M_{ij}|.$$

Every operation here is in the source real closed field, and absolute values and the finite maximum are semialgebraic functions. Set

$$E_0 = C_{00} - B, \qquad A_{pq} = \delta^{p+q} C_{pq}, \qquad r_{pq} = n(A_{pq}).$$

Our explicit source admission condition is

$$n(E_0) + \sum_{(p,q)\in I} r_{pq} \le \tfrac{1}{6}. \tag{12}$$

It is imposed together with the printed rational stationary-source formulas, native positive denominators, $D, \beta \ge 0$, and $\lambda > 0$. The condition contains no unknown testing value, no choice of a real standard part, and no unspecified sufficiently small constant.

The elementary inequality $2|x_ix_j| \le x_i^2 + x_j^2$ proves $-n(M)I \preceq M \preceq n(M)I$ over every ordered real closed field. Consequently all the following are positive semidefinite source matrices:

$$H_0 = C_{00} - \sum_I r_{pq} I \succeq \tfrac{1}{6} I, \qquad H_{pq}^{\pm} = \tfrac{1}{2}(r_{pq} I \pm A_{pq}) \succeq 0.$$

On the fixed testing rectangle $0 \le s, w \le \delta$, let $a = s/\delta$, $b_t = w/\delta$. Direct expansion gives

$$Q = H_0 + \sum_I \big[H_{pq}^{+}(1 + a^p b_t^q) + H_{pq}^{-}(1 - a^p b_t^q)\big]. \tag{13}$$

This is a matrix polynomial identity, not a positivity approximation.

A version with one numerical matrix envelope. The same dictionary also applies on the more elementary sufficient domain

$$n(E_0) \le \tfrac{1}{12}, \qquad n(C_{pq}) \le 128 \quad ((p,q) \in I).$$

Indeed, put $T_\delta = 2\delta + 2\delta^2 + \delta^3 + \delta^4 + \delta^5$. At $\delta = 2^{-12}$, rational arithmetic gives $128T_\delta < 1/12$. Replace the source matrices above by

$$H_0 = C_{00} - 128T_\delta I, \qquad H_{pq}^{\pm} = \tfrac{1}{2}\delta^{p+q}(128I \pm C_{pq}).$$

They are positive semidefinite, with $H_0 \succeq I/6$, and the same matrix identity holds. Here even the matrix envelope is a fixed numerical constant. Section 4 establishes these inequalities throughout the printed source box. They are numerical source estimates, not a consequence of pointwise boundedness alone.

Only two gate radicals. Write $a = A_t^4$, $b_t = B_t^4$, and introduce

$$\xi^2 = 1 - A_t^4, \qquad \eta^2 = 1 - B_t^4.$$

For every nonnegative integer pair (p, q),

$$\begin{aligned} 1 + A_t^{4p} B_t^{4q} &= 1 + (A_t^{2p} B_t^{2q})^2, \\ 1 - A_t^{4p} B_t^{4q} &= \sum_{i=0}^{p-1} (\xi A_t^{2i})^2 + \sum_{j=0}^{q-1} (\eta A_t^{2p} B_t^{2j})^2. \end{aligned}$$

Empty sums mean zero. The second identity is a finite geometric sum. For the seven pairs in I, the resulting common multiplier list is

$$\begin{aligned} \mathcal{M} = \{&1, B_t^2, B_t^4, A_t^2, A_t^4, A_t^4 B_t^2, A_t^8, A_t^8 B_t^2, \\ &\eta, \eta B_t^2, \eta A_t^4, \eta A_t^8, \xi, \xi A_t^2, \xi A_t^4, \xi A_t^6\}. \end{aligned}$$

In particular there is no new radical for each monomial, and no $\sqrt{M \pm y(t)}$ depending on an adaptively chosen M.

A compact algebraic cover of the testing chart. Set

$$u = \delta^2 A_t^8, \quad w = \delta B_t^4, \quad W = \omega^4, \quad Z = \zeta^4, \qquad \omega^8 + \zeta^8 + X^2 + Y^2 = 1.$$

Together with the two gate equations, these equations define a compact real algebraic testing set. They cover exactly the normalized real-state chart $W, Z \ge 0$, $0 \le \sqrt{u}, w \le \delta$. All signs of $A_t, B_t, \omega, \zeta, \xi, \eta$ are allowed. The common state sign takes the same-sign state chart $WZ \ge 0$ to this chart. In particular

$$\sqrt{uwWZ} = \delta^{3/2} A_t^4 B_t^2 \omega^2 \zeta^2$$

is the nonnegative root for every real point of the cover. Since $\delta = 2^{-12}$, every power $\delta^{j/4}$ appearing in the substitution is rational.

A primitive rescaling of the twelve testing functions is

$$\begin{aligned} \chi = (&X, Y, \zeta^4, \omega^4, A_t^8 \zeta^4, B_t^4 \zeta^4, A_t^8 \omega^4, A_t^8 B_t^4 \omega^4, A_t^{16} \omega^4, \\ &A_t^{12} \omega^4, A_t^4 \omega^4, A_t^4 B_t^2 \omega^2 \zeta^2). \end{aligned}$$

The circuit coordinates from the preceding section satisfy $u^{1/4} = \delta^{1/2} A_t^2$ and $w^{1/4} = \delta^{1/4}|B_t|$. Their squared identities remain valid when $|B_t|$ is replaced by B_t: every monomial in each root has the same parity in B_t. The same parity argument applies to ω and ζ.

The original twelve-root vector is $g = R(c)\chi$ for the printed source matrix, after multiplying its columns by

$$(1, 1, 1, 1, \delta^2, \delta, \delta^2, \delta^3, \delta^4, \delta^3, \delta, \delta^{3/2}).$$

On the negative-state chart use $Z = -\zeta^4$, delete the final coordinate of χ, and take the first eleven columns of the negative root matrix A_- from `C1_ROOT_MATRIX.json`. The exact source-to-feature map is

$$R_- = A_-[:, 1{:}11]\ \mathrm{diag}(1, 1, -1, 1, -\delta^2, -\delta, \delta^2, \delta^3, \delta^4, \delta^3, \delta).$$

Here $[:, 1{:}11]$ means the first eleven columns in one-based notation, equivalently `A_minus[:, :11]` in Python. These three negative column signs are necessary because the feature vector uses ζ^4, whereas the source root formulas use the signed state coordinate Z.

Combining repeated multipliers in (13) yields

$$g^{\mathsf{T}} Q g = \sum_{m \in \mathcal{M}} (m\chi)^{\mathsf{T}} G_m(c)(m\chi), \qquad G_m(c) \succeq 0.$$

Each G_m is a sum of congruences $R(c)^{\mathsf{T}} H R(c)$ of the explicit source matrices $H_0, H_{pq}^{\pm}$. Source-field square completion therefore gives a finite separated SOS in this one fixed testing family. Both state normals are preserved because the whole matrix Q is used, rather than a determinant or a selected restriction. For the normalized physical slack, multiply every testing function by κ_t, with

$$\kappa_t^2 (1 + \delta^4 A_t^{16})(1 + \delta^2 B_t^8) = 1.$$

The denominator is strictly positive on the entire testing set. This is a pure testing restoration, and the sign of κ_t has no effect on the Gram identity.

Evaluated rational matrices. The source files print sixteen feature blocks of length twelve, hence 192 feature slots. Exactly 152 polynomial functions are distinct. Using only the gate relations, the block products span 280 independent monomials. No state-sphere reduction is needed: every product is homogeneous of state degree two, so a relation on the normalized state chart would extend by radial scaling to an open cone and hence would already be a polynomial relation. There are 1248 symmetric Gram entries. The fully evaluated coefficient map has 1872 nonzero integer entries and exact row rank 280. Rational elimination produces an affine pencil of sixteen 12×12 blocks in 280 coefficient variables and 968 auxiliary variables. It has 2854 nonzero affine entries, with denominators only one or two. Direct multiplication verifies all 280 coefficient equations.

For precision, this evaluated pencil represents the principal Gram cone exactly: a coefficient vector is in that cone if and only if an auxiliary vector makes all sixteen blocks positive semidefinite. It is not a representation of the full quantum set. The retained circuit, event, and angular roots belong to additional fixed blocks, and a global extraction also requires source and testing coverage.

The retained circuit scalar uses the same gates. The scalar multiplying the nonnegative circuit remainder is

$$q_C = q_{C,0} - r_C u, \qquad q_{C,0} = \tfrac{1}{2} - \frac{c_A d_G}{S}\left(\frac{2}{\lambda} + k\right), \quad r_C = \frac{c_A d_G}{S} \geq 0.$$

The circuit diagonal of the principal Gram and (12) give $q_{C,0} - r_C \delta^2 \geq 1/3$. Exactly,

$$q_C = (q_{C,0} - r_C \delta^2) + r_C \delta^2 \big[\xi^2 + (\xi A_t^2)^2\big].$$

Thus multiplying the three elementary circuit roots requires only the fixed multipliers $1, \xi, \xi A_t^2$; no adaptive radical is needed here either.

6 The evaluated full-parent local source pencils

The preceding fixed multiplier construction was applied to every principal term, and all retained roots from the complete C1 identity were then restored as their own source-positive Gram blocks. Each elementary or

angular retained root has source/testing rank at most three. The circuit roots are multiplied by the three fixed functions $1, \xi, \xi A_t^2$ just derived. No guessed monomial extension is used in this compilation.

For $Z \geq 0$, the resulting fixed cone has 52 blocks, total matrix order 268, maximum block order 12, and 168 distinct testing functions. Its coefficient map has 281 independent rows and 1368 Gram entries. Substituting the full parent, linear in the seven canonical source coefficients $(d, m, C, A, D, \alpha, \beta)$, and eliminating the coefficient equations gives an evaluated rational pencil with 1087 auxiliary variables. For $Z \leq 0$, the common circuit-centre term is absent from the principal root matrix, so its testing span has eleven functions. The corresponding fixed cone has 49 blocks, total order 240, maximum block order 11, 143 distinct functions, 221 independent coefficient equations, and 934 auxiliaries.

Block-diagonal concatenation gives one evaluated rational pencil of 101 blocks and total order 508, in the same seven source coefficients and 2021 auxiliaries. Every one of its 5326 nonzero upper-triangle affine entries is printed. Direct rational multiplication verifies all 502 full-parent coefficient equations after elimination.

The equivalence proved by these matrices is exact and limited in scope: a canonical source belongs to both displayed fixed Gram cones if and only if this pencil admits a positive-semidefinite completion. Feasible Grams imply nonnegativity of its full slack on both testing state charts. Conversely, the uniform source estimates together with the explicit source Gram assignments prove membership for every admitted native source. The exact gauges

$$(u, Y, Z) \mapsto (-u, -Y, -Z), \qquad (w, X, Z) \mapsto (-w, -X, -Z)$$

preserve the full parent and extend the conclusion to $|u| \leq 2^{-24}$, $|w| \leq 2^{-12}$ and every real state. Applying the real quadratic identity to real and imaginary state parts covers complex states. The final polynomial identities specialize directly to testing axes and zero state coordinates. No state coordinate, D, or β is inverted in the final family.

These two pencils are full-state local certificates. They do not represent the full quantum set, and they do not establish that an arbitrary globally valid C1 or C2 source belongs to the same fixed cone. The uncovered source and testing regions remain a separate constructive obligation.

7 An explicit fixed family from the six-root identity

This construction represents the complete reciprocal C1 parent on an explicit source domain and an independent testing rectangle. It proves membership in a particular fixed Gram cone on that domain. It makes no assertion that the domain covers all C1 sources or any of the omitted C2 source families.

Source notation. Use (P1), (SP3), $Q = Q_b = 1 - b$, and the source quantities

$$L = \lambda/\sigma, \qquad s = \sqrt{1 + L^2}, \qquad d_G = 4D/\sigma^2.$$

Within this section s is this pure-source normalizer, not the testing square root used in the preceding section.

The exact full-state Gram. For independent testing coordinates u, w, W, X, Y, Z set

$$\begin{aligned} N_x &= X + uhZ, & N_y &= (Y - bwZ)/q, & R &= wZ/q - uqW, \\ \Gamma &= \sigma Z - uwW, & A_\theta &= q(u^2 - x)W, & \widehat{V} &= suw(u^2 - x)W. \end{aligned}$$

Let $v = (N_x, N_y, R, \Gamma, A_\theta, \widehat{V})^T$. The exact identity used here is

$$F_w = E_{\mathrm{ret}} + v^T \left(G_0 + \sum_{\gamma \in \mathcal{A}} u^{\gamma_1} w^{\gamma_2} E_\gamma \right) v, \qquad \mathcal{A} = \{00, 10, 01, 11, 20, 02, 21, 30\}. \tag{SR1}$$

Here

$$G_0 = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2b-1 & 0 & 0 & 0 \\ 0 & 2b-1 & b^2+Q^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & d_G & 0 & L/(2s) \\ 0 & 0 & 0 & 0 & 4\kappa H & 0 \\ 0 & 0 & 0 & L/(2s) & 0 & 1 \end{pmatrix}.$$

All unspecified entries of the symmetric matrices below are zero. Indices follow the six roots just printed, starting at 1.

γ	(i,j)	$(E_\gamma)_{ij}$
00	$(3,3)$	$z[8bD + 8bp\sigma + 4b\sigma(2C+2D+1)(r-1)]$
00	$(2,3)$	$2z[\sigma(2C+1)(r-1) + 2D(1+\sigma(r-1)) + 2p\sigma]$
00	$(4,6)$	$-sx/2$
10	$(1,4)$	-2ν
10	$(2,5)$	$A_y/2$
10	$(3,5)$	$(\chi + \lambda + s^2\sigma x)/2$
01	$(2,4)$	$qR_0/2$
01	$(3,4)$	$bqR_0/2$
01	$(1,5)$	$qB_x/2$
11	$(1,2)$	$-2q(2m+2D+1)$
11	$(1,3)$	$qR_x/2$
20	$(2,2)$	$2z(1+2m)$
20	$(3,3)$	zk_1
20	$(2,3)$	$2z[2Dh + 2p\sigma + (1+2m)g]$
20	$(4,4)$	L
20	$(5,5)$	$-s^2\sigma$
20	$(4,6)$	$s/2$
02	$(2,2)$	$2z(1+2D)$
02	$(3,3)$	$2zb^2(2D+1)$
02	$(2,3)$	$2zb(2D+1)$
21	$(2,4)$	$-4pq$
21	$(3,4)$	$-4bpq$
30	$(3,5)$	$-s^2\sigma/2$

(SR2)

In particular, the $N_x\Gamma$, $N_y\Gamma$, and $R\Gamma$ entries retain both complete state-normal couplings. No determinant substitution has replaced the four-state parent.

For completeness, define $X_e = X + ewW$, $J_C = Y/q - buqW$, and $J_\ell = (Y - wZ)/q + QuqW$. The retained form is

$$\begin{aligned} E_{\mathrm{ret}} &= 4\beta N_x^2 + (4\beta + 2 + 4m)(wN_x)^2 + 2(2D+1)(uX_e)^2 + 4bpz(u\Gamma)^2 \\ &\quad + [4a + 2z(2C+1) - 1]J_C^2 + (4\ell - 1)J_\ell^2 \\ &\quad + 4a(uJ_C)^2 + (4a + 4Cz)(wJ_C)^2 + 4a(uwJ_C)^2 + 4\ell(uJ_\ell)^2. \end{aligned} \tag{SR3}$$

All ten coefficients are nonnegative when $4a, 4\ell \geq 1$.

To prove (SR1), start with the full-parent identity SP6. With $V = uw(u^2 - x)W$, direct multiplication gives

$$\begin{aligned} K_A A_\theta^2 + \mathscr{S} &= 4\kappa H A_\theta^2 + Lu^2\Gamma^2 + L\Gamma V + \lambda u A_\theta R, \\ 0 &= V^2 - \sigma u^2 A_\theta^2 - u\sigma(u^2 - x)A_\theta R + (u^2 - x)V\Gamma. \end{aligned}$$

Add $s^2 = 1 + L^2$ times the second identity, replace V by $\widehat{V}/s$, and collect the eight testing monomials. Clipping the two radial squares at one gives (SR3). This proves (SR1)–(SR3) algebraically at either sign of λ.

Explicit source guards. For a real symmetric matrix set $\|T\|_{\mathrm{row}} = \max_i \sum_j |T_{ij}|$. Choose the fixed rational constants

$$\delta = 2^{-12} = 1/4096, \qquad \eta = 1/56.$$

The source admission conditions, in addition to the native domain, are

$$4a \geq 1, \quad 4\ell \geq 1, \quad d_G \geq 1, \quad 4\kappa H \geq 1, \tag{SR4}$$

$$\|E_{00}\|_{\mathrm{row}} \leq 1/12, \qquad \delta^{|\gamma|}\|E_\gamma\|_{\mathrm{row}} \leq 1/56 \quad (\gamma \in \mathcal{A} \setminus \{00\}). \tag{SR5}$$

These are finite, explicit inequalities in source-only rational and algebraic functions; no testing point is used to select them.

The radial block has determinant one and trace at most three, so it is at least $I_2/3$. An explicit pure-source proof is

$$(N_y + bR)^2 + (N_y - QR)^2 = \frac{N_y^2 + R^2}{3} + \frac{5}{3}\left(N_y + \frac{3(2b-1)}{5}R\right)^2 + \frac{1 + 6bQ}{15}R^2.$$

The $(\Gamma, \widehat{V})$ block is at least $I_2/2$: its diagonals are at least one and $|L/s| \le 1$. Thus $G_0 \succeq I_6/3$. Consequently

$$B = G_0 + E_{00} - I_6/8 \succeq I_6/8.$$

For the seven nonconstant monomials put

$$y_\gamma = (u/\delta)^{\gamma_1}(w/\delta)^{\gamma_2}, \qquad T_\gamma = \delta^{|\gamma|}E_\gamma.$$

On the fixed independent testing box $0 \le u, w \le \delta$ one has $0 \le y_\gamma \le 1$. The exact matrix identity is

$$G_0 + \sum_\gamma u^{\gamma_1}w^{\gamma_2}E_\gamma = B + \frac{1}{2}\sum_{\gamma\neq 00}\{(\eta I_6 + T_\gamma)(1+y_\gamma) + (\eta I_6 - T_\gamma)(1-y_\gamma)\}. \tag{SR6}$$

The scalar equality $7\eta = 1/8$ proves the constant coefficient. The source matrices $\eta I_6 \pm T_\gamma$ are positive semidefinite by (SR5). All factors $1 \pm y_\gamma$ are nonnegative in the testing field. Factor each source matrix over its own real closed field. This proves a separated tensor SOS with the *fixed* fourteen testing radicals $\sqrt{1 \pm y_\gamma}$. No adaptive constant M or source-dependent testing function occurs.

A full uniform source rectangle for the nonpositive branch. The guards hold for every native source satisfying

$$0 < x, \sigma \le \varepsilon := 2^{-10} = 1/1024, \qquad 0 < z \le 1, \qquad \lambda \le 0. \tag{SR7}$$

No inverse of β occurs, so its literal zero face is included. The proof below forces $D > 0$ on this nonpositive branch; the $D = 0$ source face is therefore assigned to the complementary positive branch, rather than being cancelled or omitted.

Here are elementary bounds establishing the claim. From the definition of p, $P \ge 1$ and $0 < E < 1$, one has $p \le 1 + \varepsilon < 2$. Since every term preceding $-4bp\sigma z$ in λ is nonnegative,

$$H \le 2, \quad h \le 2\sigma, \quad e \le 2x, \quad \beta\theta^2 \le 2\sigma^2, \quad -8 \le L \le 0, \quad 1 \le s \le 9.$$

In particular $\theta \ge 1/2$ and $\beta\theta \le 4\sigma^2$. The source formulas give

$$1 \le P \le 1 + 4\varepsilon, \quad 1 - \varepsilon \le M \le 1 + \varepsilon, \quad 1 - \varepsilon \le J \le 1 + \varepsilon, \quad B_s = P + xz(J - \sigma) \le 1 + 6\varepsilon.$$

The exact identity $2\theta P\beta = hB_s - \sigma H_0$, where $H_0 = 1 + zQ \ge 1$, implies $H \ge 3/4$. Also

$$3/8 \le \kappa \le 1, \quad 1/4 \le p \le 2, \quad 1/4 \le l \le 1, \quad C(1-r) \le 1, \quad m \le 1/8.$$

For the lower bound on l, use $2Pl \ge M(1-e) \ge (1-\varepsilon)(1-2\varepsilon)$. Its upper bound follows from $eR_s(1+\sigma z) + M(1-e) \le (1+\varepsilon)^2$. These estimates already prove the first, second and fourth guards in (SR4).

To bound D, use the exact BE4 identity

$$2P(JD + x\theta\beta) = \sigma^2\mathcal{H},$$

where

$$\mathcal{H} = R_sH_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + Ke^2\}.$$

The preceding inequalities give $1 - 5\varepsilon \le \mathcal{H} \le 3$, and hence $D \le 3\sigma^2$. Moreover

$$d_G \ge \frac{2(1-5\varepsilon)/(1+4\varepsilon) - 16\varepsilon}{1+\varepsilon} \ge 1;$$

the last inequality follows from $1 - 31\varepsilon - 68\varepsilon^2 \ge 0$. Thus (SR4) is proved.

The remaining coefficient estimates are deliberately loose:

$$|\nu| \le 3, \quad |R_0| \le 29, \quad |A_y| \le 22, \quad |\chi| \le 24, \quad |R_x| \le 6.$$

They follow by substituting the displayed bounds in the five explicit source formulas. For example $|g| \le 1$ and $pb + lQ \le 2$; all appearances of C in (SR2) outside whole positive squares occur as $C(1-r)$, whose bound is one. Direct row sums in (SR2) then give

$$\|E_{00}\|_{\mathrm{row}} \le 64\varepsilon, \qquad \|E_\gamma\|_{\mathrm{row}} \le 64 \quad (\gamma \neq 00).$$

For the constant row, the sharper estimates are $|(E_{00})_{33}| + |(E_{00})_{23}| \le 42\sigma + 12D(1+\sigma) < 43\varepsilon$ and $|(E_{00})_{46}| \le 9x/2$. Thus $64\varepsilon = 1/16 < 1/12$ and $64\delta^{|\gamma|} \le 1/64 < 1/56$, proving (SR5). These are finite real inequalities, so they hold over arbitrary real closed source and testing fields by the same algebraic proof.

The fixed testing span. Set

$$\phi_{10} = (W, X, Y, Z, uZ, wZ, uW, uwW, u^2W, u^3wW)^T.$$

The exact 6×10 source matrix T satisfying $v = T\phi_{10}$ is

$$T = \begin{pmatrix} 0 & 1 & 0 & 0 & h & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/q & 0 & 0 & -b/q & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/q & -q & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma & 0 & 0 & 0 & -1 & 0 & 0 \\ -qx & 0 & 0 & 0 & 0 & 0 & 0 & 0 & q & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -sx & 0 & s \end{pmatrix}.$$

The first principal block is $T^T BT$; the other fourteen principal blocks are

$$\tfrac{1}{2} T^T(\eta I_6 \pm T_\gamma)T \quad (\gamma \neq 00)$$

on testing vectors $\sqrt{1 \pm y_\gamma}\,\phi_{10}$. All these matrices are positive semidefinite by (SR4)–(SR6).

The independent algebraic testing cover adds fourteen variables $\zeta_{\gamma,\pm}$ with

$$\zeta_{\gamma,\pm}^2 = 1 \pm y_\gamma, \qquad b_u^2 = u(\delta - u), \qquad b_w^2 = w(\delta - w),$$

and the equations

$$W^2 + X^2 + Y^2 + Z^2 = 1, \qquad \kappa_t^2(1 + u^2)(1 + w^2) = 1.$$

These eighteen equations in twenty-three real variables define a compact algebraic set covering the entire test rectangle and state sphere. Multiply every testing vector by κ_t to obtain the normalized physical slack. Each radical multiplies a whole Gram block, so either sign of every radical is allowed. Axes and zero state coordinates are literal specializations. The gauges (SG) and real/imaginary splitting cover negative angles and complex states.

Preserving the elementary root structure: the primary pencil. Keep all fifteen principal blocks on ϕ_{10}. Absorb the retained N_x, J_C, J_ℓ squares into the first principal block; these three rows already lie in span ϕ_{10}. Group the seven remaining retained rows only as follows:

fixed testing vector	retained roots	block order
(wX, uwZ)	wN_x	2
(uX, wY, uwW)	$uX_e,\ wJ_C$	3
(uZ, uwY, u^2wW)	$u\Gamma,\ uwJ_C$	3
(uY, uwZ, u^2W)	$uJ_C,\ uJ_\ell$	3

Each block is the sum of the nonnegative source multiples of the displayed coefficient-vector outer products. Thus every block is positive semidefinite over the source field. This compilation has exactly nineteen blocks, fifteen of order ten and four of orders $(2, 3, 3, 3)$, with total order 161 and maximum block order ten. It preserves the testing support of each existing root rather than adding arbitrary cross terms among all seventeen base monomials.

The evaluated rational source pencil is `six_root_structured_source_pencil.coo.json`. Its visible coordinates are $(d, m, C, A, D, \alpha, \beta)$; it has 666 free auxiliary coordinates and 2311 nonzero upper-triangular rational coefficient records. The common feature-product expansion has 180 coefficient equations and 846 raw upper-triangular Gram variables. Exact elimination has rank 180, leaving $846 - 180 = 666$ free variables. The matching map and every resulting source and auxiliary coefficient are supplied and exactly checked.

Here both directions of the local cone representation are algebraic. An affine-pencil witness supplies positive semidefinite Gram blocks; the checked coefficient map proves their weighted sum equals the complete parent. Conversely, any collection of Gram blocks giving that parent is a solution of the 180 printed linear equations. Take its entries at the printed free positions as the 666 auxiliaries. The exact pivot equations then reconstruct every other entry, so the printed affine pencil represents those same Gram blocks. Thus this pencil is exactly the specified local fixed Gram cone. This is a two-sided statement about that cone, not a quantum reverse-inclusion claim.

An admitted rational source for the general guards is $\sigma = 10^{-4}$, $x = 10^{-8}$, $b = 1/2$, $z = 1$, $h = 3/20000$. It has $\lambda > 0$ and is therefore not an example from the $\lambda \leq 0$ subdomain. The nineteen source Gram matrices at this point are printed in the structured fixture, with 331 nonzero rational upper-triangle entries. The

source radical s cancels from their evaluated entries. This fixture is an additional exact consistency check; the uniform admission theorem is proved by the preceding inequalities.

The result is a fixed local testing cone with a proved admission theorem. It is not a primal affine lift of the full eight-coordinate set. To obtain such a lift one still needs a finite cover of every valid source and every independent test by compatible fixed testing families, including the remaining C1 and C2 boundaries.

8 Exact affine compilation and both local inclusions

For each fixed chart, write the fixed testing blocks as $f_b(t)$. Expanding

$$\sum_b f_b(t)^{\mathsf{T}} G_b f_b(t) = F_{\mathbf{c}}(t) \tag{CM}$$

and reducing the explicitly printed gate equations gives a rational linear system

$$\mathcal{A}g = \mathcal{B}\mathbf{c}.$$

Here g lists the upper-triangular entries of all symmetric G_b. An off-diagonal Gram entry contributes twice its coefficient to (CM); the matrix file stores the actual entry, not this doubled coefficient. All coefficients are rational because $\delta = 2^{-12}$ has rational quarter powers.

The positive chart has 1368 Gram variables and 281 independent equations. The negative chart has 1155 variables and 221 independent equations. The structured six-root cone has 846 variables and 180 independent equations. The coefficient maps, ordered monomials, pivots, and every affine entry are supplied. Exact rational elimination expresses the pivot entries as affine functions of $\mathbf{c}$ and the remaining 1087, 934, or 666 free entries, respectively. The files verify these expressions by substitution into all the original coefficient equations.

This gives both directions without any positivity inference from a determinant. If the affine pencil is positive semidefinite, its blocks are feasible Grams in (CM), hence give a fixed-family certificate on the entire chart. Conversely, any fixed-family certificate supplies a solution g of (CM). Choose its free entries as the auxiliary coordinates. The pivot equations reconstruct its remaining entries exactly, so the affine pencil equals those same positive semidefinite blocks.

Because the two state charts must both be certified, their pencils share the visible coordinates and are intersected:

$$L_{508}(\mathbf{c}, y_+, y_-) = \operatorname{diag}(L_{\mathrm{POS}}(\mathbf{c}, y_+), L_{\mathrm{NEG}}(\mathbf{c}, y_-)).$$

The two circuit-sign *constructions* are alternative sufficient certificates on their common testing box. We include both by a source-cone sum. With a free split vector $z_c \in \mathbb{R}^7$ put

$$L_{669}(\mathbf{c}, z_c, y_+, y_-, y_6) = \operatorname{diag}\big(L_{508}(z_c, y_+, y_-), L_{161}(\mathbf{c} - z_c, y_6)\big). \tag{LP}$$

This proves Theorem 2 from the definition of $\mathcal{K}_{669}$. The split is not a positivity-constrained source parameter: it is simply the visible vector of the first certificate summand.

For Theorem 1, use $z_c = \mathbf{c}$ and the zero witness in the six-root cone when $\lambda > 0$. The explicit twelve-root source assignment, numerical reserve, and retained squares provide both state-chart witnesses. If $\lambda \le 0$, use $z_c = 0$ and the six-root source assignment. The larger six-root bound $x, \sigma \le 2^{-10}$ contains (D). For every feasible vector in the sum, add the two full-parent identities. Their common testing rectangle is (T), and each identity is nonnegative there. Apply (SG) and homogeneity in the state to finish the proof.

Reading the actual matrices. The primary file uses the tuple

$$[\, b, i, j, v, a \,]$$

to mean that rational number a multiplies coordinate v in entry (i, j) of block b. All indices are zero-based; $i \le j$. Mirror an off-diagonal entry once. Coordinates $0, \dots, 6$ are $(d, m, C, A, D, \alpha, \beta)$. The following 2694 coordinates are listed in `auxiliary_order`; their first seven are the components of z_c. There is no constant matrix. Thus the file is directly an evaluated collection of rational matrices

$$L_{669}(\mathbf{c}, y) = \sum_{j=0}^{6} c_j A_j + \sum_{k=0}^{2693} y_k B_k.$$

The source formulas and algebraic roots are used to *prove* feasibility on (D), not to compute the entries of A_j or B_k.

Exact checks supplied with the proof. The scripts independently check the generic P1-to-P4 parent identity, both conductor identities, all retained elementary, angular, and circuit roots at both state signs, and the final rational coefficient equations. The last check can be run using only Python's standard library; symbolic regeneration additionally uses SymPy. No archived checker is imported, and no numerical optimization or CHSH comparison is a premise of either theorem. The negative-state source congruence has the explicit signed column scaling printed in Section 5.

9 What is still missing for the full eight-coordinate set

The local result resolves a concrete part of the constructive problem: pointwise bounded-residue arguments have been replaced by one fixed finite algebraic family on an explicit source and independent testing box. This does not supply the global Bell slack cone.

A full extraction still needs a finite algebraic testing cover $\pi : \mathcal{T} \to \mathcal{P}$ onto every physical two-qubit testing strategy, together with fixed testing vectors f_b, for which every valid affine Bell slack satisfies an identity

$$\ell_{\mathbf{a}}(\pi(t)) = a_0 + \sum_{\nu=1}^{8} a_\nu \Phi_\nu(\pi(t)) = \sum_b f_b(t)^{\mathsf{T}} G_b(\mathbf{a}) f_b(t), \qquad G_b(\mathbf{a}) \succeq 0. \tag{GLOBAL}$$

The assignment $\mathbf{a} \mapsto G_b(\mathbf{a})$ need not be affine; the coefficient equations in the Gram variables must be affine. The fixed functions and their algebraic domains must be independent of $\mathbf{a}$. A finite disjoint union of testing charts is allowed, provided every chart is checked and the resulting fixed cone contains every valid slack.

For the present construction the uncovered obligations are the other source regimes, including $z > 1$ and the other C1/C2 branches; literal source openings $x = 0$, $\sigma = 0$, or $z = 0$; and independent tests outside (T). The state-coordinate zeros and both state signs *within* (D)–(T) have already been handled. A local positive reserve cannot be extended across an uncovered source or testing boundary by merely asserting compactness.

If (GLOBAL) is proved with evaluated finite product maps, physical evaluation gives the forward inclusion of its moment lift. For the reverse inclusion, the same Gram identity makes every valid Bell slack nonnegative on every feasible projected moment vector. A point outside the closed quantum set has a separating valid affine slack and is therefore excluded. This is the missing universal adequacy argument; a determinant identity, one stationary neighbourhood, or agreement on CHSH does not replace it.

Why collecting the old TF1 radicals is not a finite solution. For distinct real numbers $M_1, \dots, M_N$, the square classes of $y + M_i$ are independent in $\mathbb{R}(y)^\times/\mathbb{R}(y)^{\times 2}$: the valuation at $y = -M_i$ is odd in precisely the ith factor of any nonempty product. Hence

$$[\,\mathbb{R}(y)(\sqrt{y+M_1}, \dots, \sqrt{y+M_N}) : \mathbb{R}(y)\,] = 2^N.$$

A fixed finite algebraic function field cannot contain all adaptive choices. This is an obstruction to keeping the original radicals unchanged, not a nonexistence result for another fixed SOS representation. The two-gate construction above is such a different representation on the explicit box.

Pointwise boundedness also gives no uniform real rectangle without further estimates. For infinitesimal x and any bounded testing element y, the scalar $1 + x \otimes y$ has positive real residue. But every ordinary rectangle $0 < x \le \epsilon$, $y \in \mathbb{R}$ contains points where $1 + xy < 0$. The numerical source inequalities proved here are therefore an essential additional step, not a consequence of a rank bound alone.

No claim about the reverse inclusion of the existing order-13744 or order-63888 candidates follows from this construction. The new evaluated matrices represent the local cone stated in Theorem 2, and the full-set extraction remains open.

Primary source and reproducibility

The source identities are those of `latex/Uniform_C1_Certificate.tex` and the complete companion in the public repository

`https://github.com/anubhav70747/q222-finite-sdp`

at commit `0c1ce2e15ed067a6845464e2468559cea5274f2c`. The present note supplies the evaluated local constants, fixed families, and rational elimination in addition to those identities. The source bundle contains the proof's LaTeX, all fixed features, coefficient maps, rational pencils, exact verifiers, and input hashes. Its README gives one-command verification and regeneration instructions.